

Nneka Asuzu

LinkedIn: <https://www.linkedin.com/in/nneka-asuzu/>

GitHub: <https://github.com/NnekaAsuzu>

Portfolio: <https://nnekaasuzu.github.io/>

PROFESSIONAL SUMMARY

Data Scientist with experience applying machine learning and statistical modeling to support business decision-making and operational efficiency. Skilled in predictive modeling, segmentation, forecasting, and analytical reporting across multiple business domains. Experienced in data preparation, feature engineering, model evaluation, and translating analytical findings into actionable business insights.

CORE SKILLS

Programming & Analytics: Python (Pandas, NumPy, Scikit-Learn), SQL, Excel

Machine Learning & Statistical Modeling: Regression, Classification, Random Forest, XGBoost, Gradient Boosting, Clustering, PCA, Feature Engineering, Model Evaluation (ROC-AUC, F1-score, RMSE)

Statistics & Experimentation: A/B Testing, Hypothesis Testing, Confidence Intervals, Bootstrap Analysis

Data Visualization & BI: Power BI, Tableau, Plotly Dash, Dashboards, Data Storytelling

Tools & Platforms: Azure ML, Git, MLflow, Jupyter

RELEVANT WORK EXPERIENCE

Data Scientist | SprintLab Digital – Toronto, Ontario | Jan 2023 – Present

Consulting firm delivering machine learning and data modeling solutions across multiple business domains.

- Develop and tune machine learning models (Random Forest, XGBoost, Gradient Boosting) to improve prediction accuracy and support decision-making across client projects.
- Analyze and transform structured datasets using Python and SQL to support predictive modeling, segmentation, and business analytics.
- Evaluate classification and regression models using ROC-AUC, precision-recall, RMSE, and business KPIs to guide model selection.
- Deliver data-driven insights through dashboards, reporting solutions, and stakeholder collaboration to support business performance.

Data Scientist | Teleperformance – Toronto, Ontario | Feb 2021 – Dec 2022

Global leader in customer experience management, supporting operations across multiple departments.

- Developed machine learning models (Random Forest, XGBoost) for customer behavior prediction and segmentation, improving outcomes by 12–15%.
- Applied Python (Pandas, Scikit-Learn) and SQL to prepare and analyze operational datasets for predictive modeling and analytics.
- Built statistical and predictive models for workforce and operational optimization, improving efficiency by 8–10%.
- Designed dashboards and reporting workflows to support operational monitoring and decision-making.

SELECTED PROJECTS

Credit Risk Prediction Model

GitHub: https://github.com/NnekaAsuzu/credit_risk_loan_default_prediction

- Built classification models (XGBoost, Random Forest) to predict loan default risk on financial datasets.
- Engineered financial, credit, income, and behavioral features to improve predictive performance.
- Evaluated models using ROC-AUC, F1-score, and precision-recall metrics.
- Applied SHAP explainability techniques to identify key drivers of loan default risk and support model interpretability.
- Delivered analytical insights and risk segmentation outputs to support lending and portfolio decision-making.

Supply Chain Risk Prediction & Inventory Optimization

GitHub: https://github.com/NnekaAsuzu/supply_chain_risk_prediction

- Developed machine learning models (Logistic Regression, XGBoost, AutoML benchmarking) to predict supplier delivery risk and support inventory optimization.
- Engineered predictive features from operational and supplier data, including lead time variability and reliability metrics.
- Evaluated models using F1-score and AUC to improve classification performance and model selection.
- Delivered supplier risk insights through Power BI dashboards to support inventory planning and operational monitoring.

Customer Value & Lifecycle Modeling (CLV, Segmentation & A/B Testing)

GitHub: https://github.com/NnekaAsuzu/customer_value_lifecycle_modeling

- Engineered customer behavioral features (recency, frequency, monetary value) using SQL and Python.
- Applied K-Means clustering and PCA for customer segmentation across High-Value, At-Risk, and Low-Value customer groups.
- Built XGBoost regression models to predict customer lifetime value (CLV) and support retention analysis.
- Designed A/B testing simulations to evaluate retention strategies and support customer engagement decision-making.

EDUCATION

Master of Science in Management and Systems

New York University, NY

Specialization: Database Technologies, Data Analytics & Applied Data Science

Bachelor of Science, *cum laude*

Arizona State University, AZ

Dual Majors: Business Management & Business Administration

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

- Data Science & Machine Learning – University of Toronto, Data Sciences Institute
- Data Analytics with Python – Ontario Tech University
- Data Analytics & Reporting with Power BI – Toronto Innovation College
- Cloud Computing & Data Science Certificate – Seneca Polytechnic